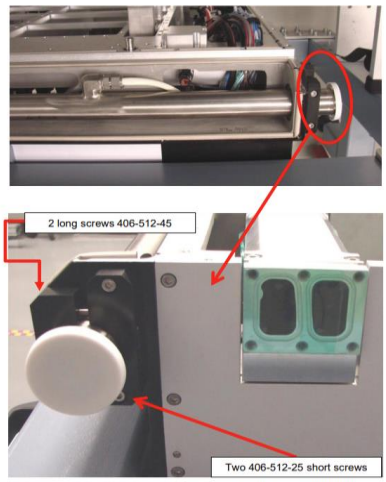
Build Overview:

1 **Return QD Assembly Install**
2 **Return Manifold Install**
3 **Inlet Manifold Install**



CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
Manufacturing Engineer

Checked By:
IVAN NOWELL PANIQUE
Team Lead

Approved By:
SHERWIN ARIOSTE
Operations Manager

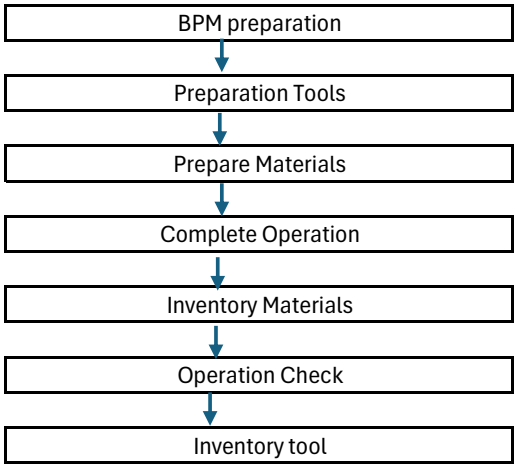
MAN: CERTIFIED TECHNICIANS.

MATERIAL:

NO.	PART NO.	DESCRIPTION	QTY.
1	638-030-00	DOW CORNING MOLYKOTE 111 COMPOUND GRAY 6G PACKAGE	1
2	361-607-06	O-RING 1-1/2IDX 1/16W E 0540-80	1
3	640-625-00	ALUMINUM MANIFOLD 12 SLOT PIPE BRACKET OR MISC	1
4	406-512-12	SCRW CP SKT GD M5X.8P/12 MM SS W/NYLOK	10
5	639-733-00	PLASTIC, CONDUCTIVE ELASTOMER GASKET, SEAL OR GASKET	1
6	633-061-01	ALUMINUM PIPE CLAMP BRACKET OR MISC ARTICLE	1
7	406-512-25	SCRW CP SKT HD M5X.8P/25 MM SS W/NYLOK	2
8	406-512-45	SCRW CP SKT HD M5X.8P/25 MM SS W/NYLOK	2
9	653-835-01	ELEC-MECH DB12 RETURN FLEX TUBE W/ QD ASSEMBLY	1
10	361-607-05	O-RING 1-3/8IDX 1/16W E 0540-80	2
11	629-106-00	CAST STEEL OR IRON MANIFOLD 24 SLOT END INLET MISC	1

METHOD:

Flow chart of operation steps



TOOL NAME	TORQUE VALUE
4mm ball head torque wrench	3.3 N.m
4mm hex bit torque wrench	3.3 N.m
11/16 open-end wrench	NA
11/16 long sleeve torque wrench	27 N·m
adjustable wrench	NA

Matters needing attention

1. Do ESD and protective safety measures.
2. Be careful during the assembly process so as not to scratch the surface of the parts.
3. After assembly, check the assembled parts for defects and leak-free operation.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

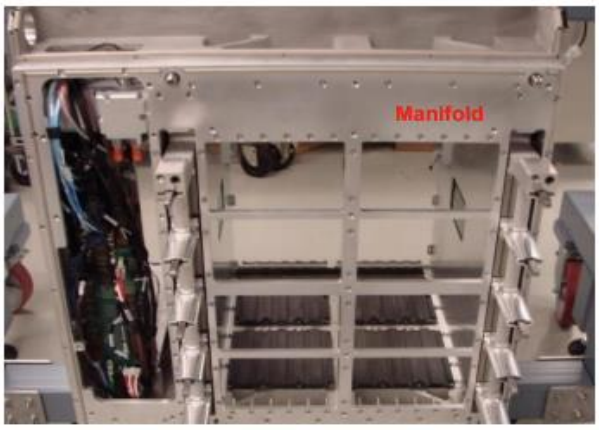
MACHINE: N/A

METHOD:

Operation steps

1.1 Rotate the T/H direction as shown to find the Manifold exit.

1.2 Carefully remove the plastic film on the Manifold outlet as shown in the figure, and check the circular groove for damage and debris.



MAN: CERTIFIED OPERATOR, TECHNICIAN

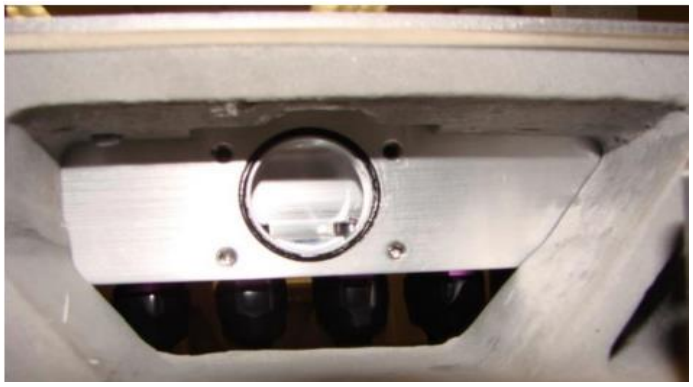
ITEM	PART NUMBER	DESCRIPTION	QTY
1	638-030-00	DOW CORNING MOLYKOTE 111 COMPOUND GRAY 6G PACKGE	1
2	361-607-06	O-RING 1-1/2IDX 1/16W E0540-80	1
3	640-625-00	ALUMINUM MANIFOLD 12 SLOT PIPE BRACKET OR MISC ARTICL	1
4	406-512-12	SCRW CP SKT HD M5X.8P/12 MM SS W/NYLOK	4

MACHINE:

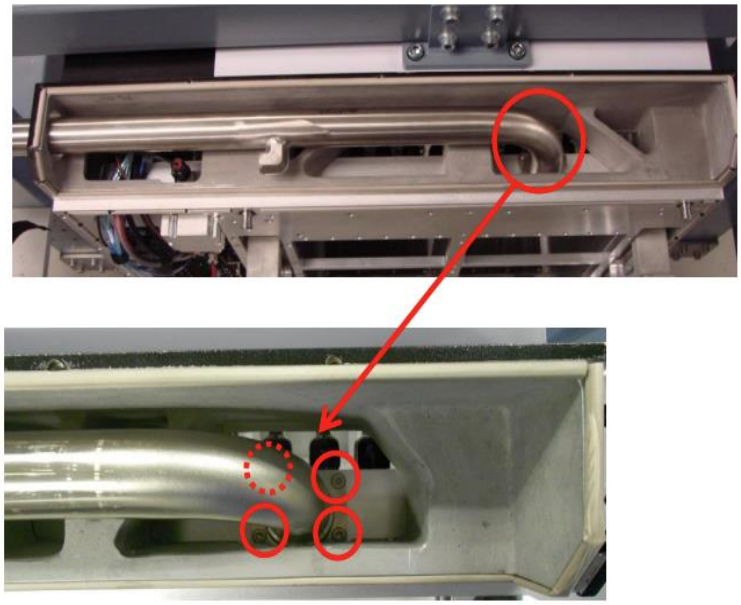
4mm ball head torque wrench 3.3 N·m

METHOD:

1.3 Use Molykote grease (638-030-00) to lubricate the O-ring (361-607-06). Put the lubricated O-ring into the outlet of Manifold.



1.4 Take a tube (640-625-00) and place it on the T/H frame as shown. Use 4 M5 screws (406-512-12) to fix the tube, The tool is a 4mm ball head torque wrench with a torque value of 3.3 N·m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

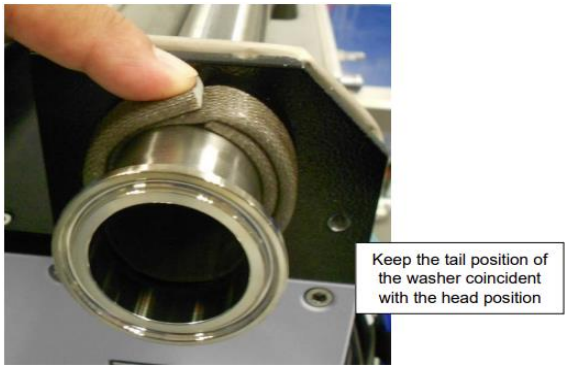
ITEM	PART NUMBER	DESCRIPTION	QTY
1	639-733-00	PLASTIC,CONDUCTIVE ELASTOMER GASKET,SEAL OR GASKET	1

MACHINE:

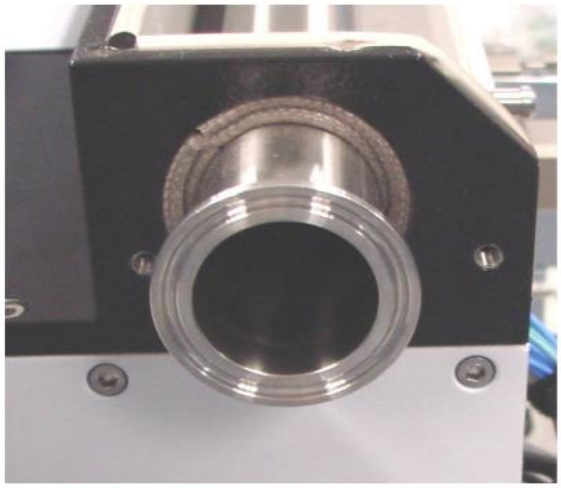
N/A

METHOD:

1.5 Install a washer (639-733-00) around the pipe as shown in the figure to ensure that the tail position and the head position remain the same after the washer is wound twice.



1.6 Place the washer into the T/H frame as shown in the figure to ensure that the washer is fully pressed into the ring when it is inserted.



MAN: CERTIFIED OPERATOR, TECHNICIAN

ITEM	PART NUMBER	DESCRIPTION	QTY
6	633-061-01	ALUMINUM PIPE CLAMP BRACKET OR MISC ARTICLE	1
7	406-512-25	SCRW CP SKT HD M5X.8P/25 MM SS W/NYLOK	2
8	406-512-45	SCRW CAP SKT HD M5X.8P/45 MM L SS W/NYLOK	2

MACHINE:

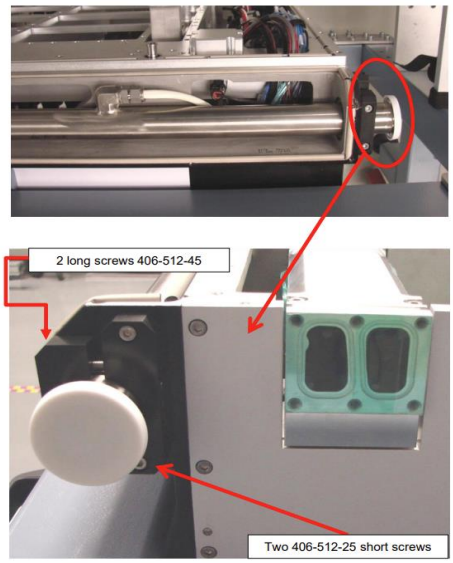
4mm ball head torque wrench 3.3 N·m

METHOD:

1.7 Install a Manifold tube clamp (633-061-01). This is composed of 2 clips, remove the original screw and throw it away.

a. Install 2 clips as shown in the picture, use 2 M5 screws (406-512-25) and 2 M5 X 45 screws (406-512-45) to fix, The tool is a

b. Place the plastic cap on the tube.



MAN: CERTIFIED OPERATOR, TECHNICIAN

ITEM	PART NUMBER	DESCRIPTION	QTY
1	653-835-01	ELEC-MECH DB12 RETURN FLEX TUBE W/ QD ASSEMBLY FOR S	1

MACHINE:

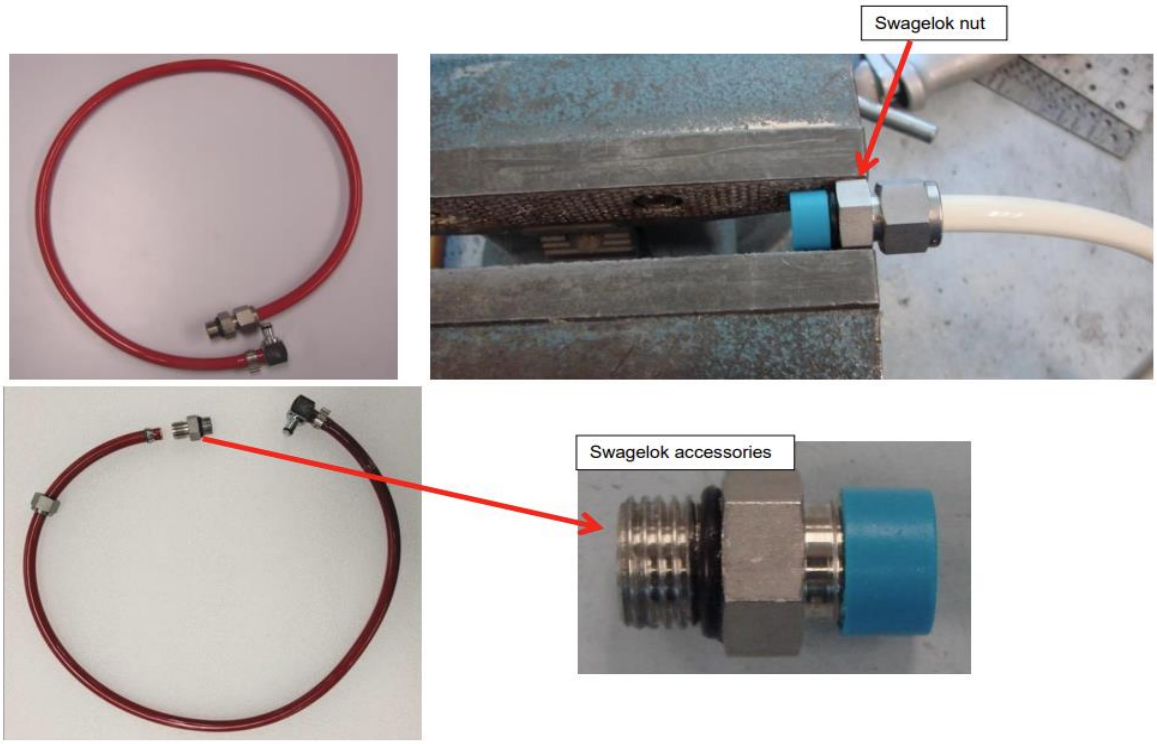
11/16 open-end wrench

METHOD:

1.8 Take one QD assembly 653-835-01.

a. Place the QD assembly on the vise and clamp it on the plane of the Swagelok nut, as shown in the figure below.

b. Use an 11/16 open-end wrench to turn the joint nut, loosen and remove the HFE hose.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: 11/16 long sleeve torque wrench 27

METHOD:

1.9 Remove the Swagelok accessories from the previous steps and remove the end ca

a. Use Molykote grease (638-030-00) to lubricate the thread and O-ring on the joint sleeve before installation.

b. Install the joint sleeve on the pipe and tighten it manually. Make a mark at the joint sleeve and elbow. Use an 11/16 long sleeve torque wrench to fix,

The torque value is 27 N·m.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

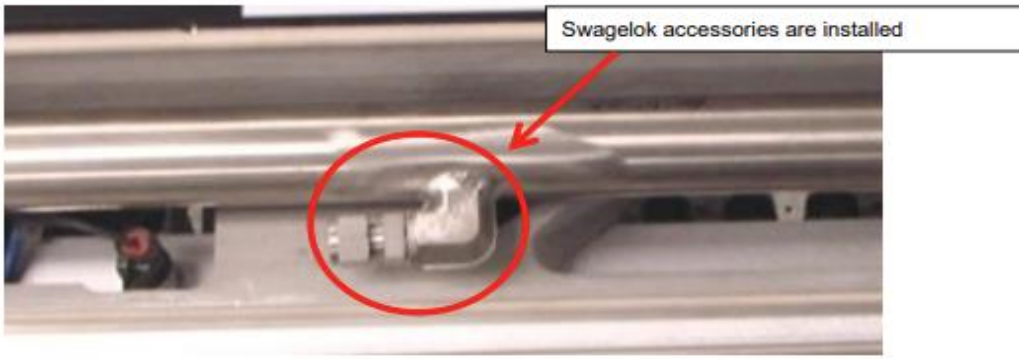
MACHINE: adjustable wrench

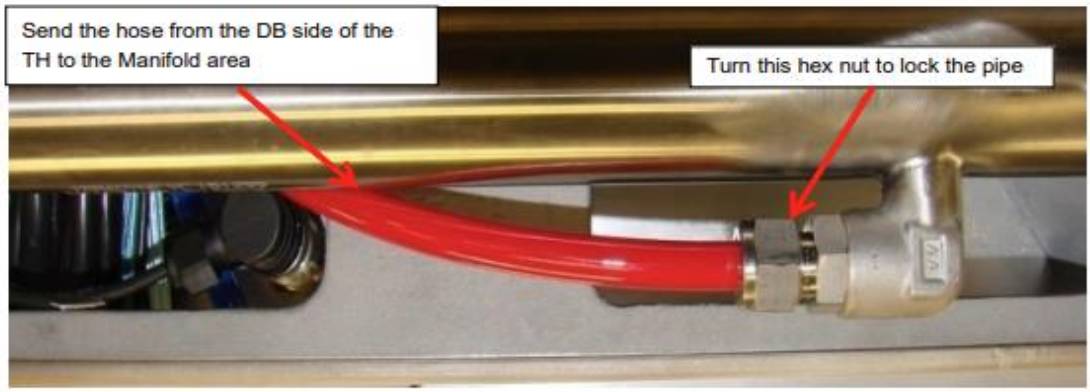
METHOD:

c. Send the red hose from the DB side of the TH to the Manifold area of the back plate end.

d. Install the hose into the Swagelok connector and push it until it stops.

Tighten the joint nut finger, and then use an adjustable wrench to turn the joint nut 1/2 turn to tighten.





MAN: CERTIFIED OPERATOR, TECHNICIAN

ITEM	PART NUMBER	DESCRIPTION	QTY
10	361-607-05	O-RING 1-3/8IDX 1/16W E0540-80	2

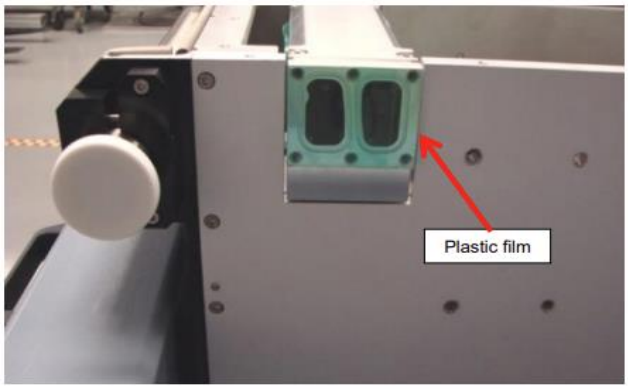
MACHINE:

N/A

METHOD:

2.1 Carefully remove the T/H plastic protective film as shown, and check that the two O-ring grooves are not damaged and debris is on the surface.

2.2 Before installing 2 O-rings (361-607-05), use Molykote grease (638-030-00) to lubricate the O-rings.



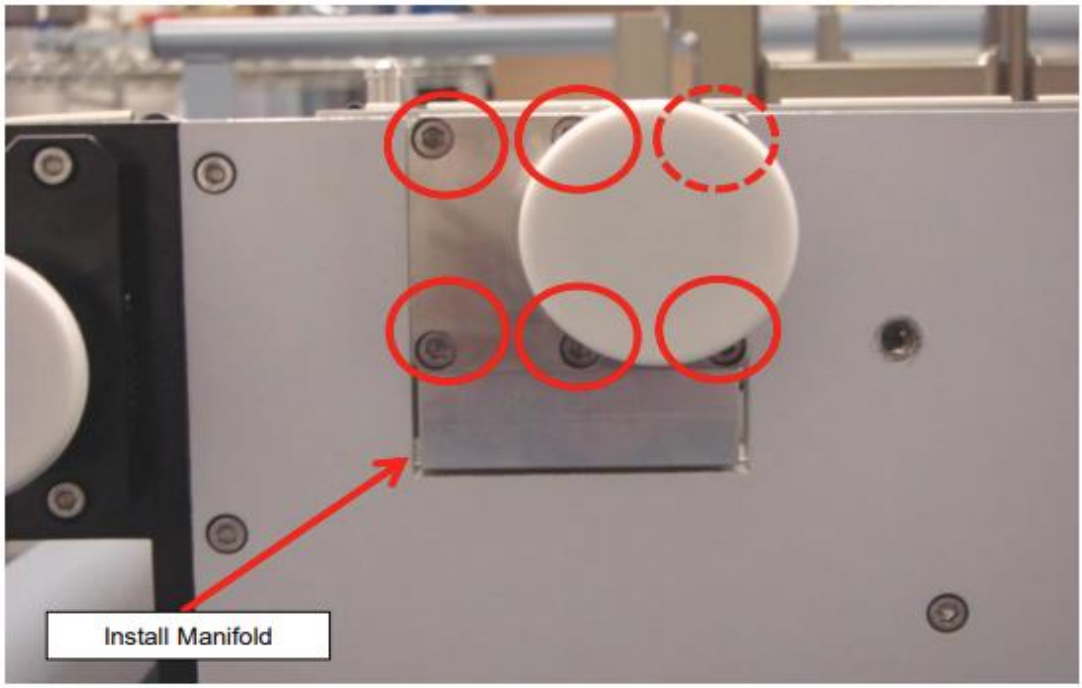
MAN: CERTIFIED OPERATOR, TECHNICIAN

ITEM	PART NUMBER	DESCRIPTION	QTY
11	629-106-00	CAST STEEL OR IRON MANIFOLD 24 SLOT END INLET MISC	1
12	406-512-12	SCRW CP SKT HD M5X.8P/12 MM SS W/NYLOK	6

MACHINE:

4mm hexagonal head torque wrench 3.3 N·m

METHOD:

2.3 Install 1 Manifold (629-106-00), fix it with 6 M5 screws (406-512-12), the tool is a 4mm hexagonal head torque wrench, The torque value is 3.3 N·m. Put on the plastic cover.

2.4 Note that when installing Manifold, the nozzle direction should be on the right.